

Module–1(Fundamental)

Q. What is SDLC Software development life cycle ?

Ans : software development life cycle is the process of planning, writing, modifying, and maintaining software. Developers use the methodology as they design and write modern software for computers, cloud deployment, mobile phones, video games, and more. Adhering to the SDLC methodology helps to optimise the outcome.

Q. What is software testing?

Ans : Software testing is a process used to identify correctness, completeness and quality of developed computer software.

Q. What is agile methodology?

Ans : Agile methodology is a combination of iterative and incremental model with focus on process adaptability and customer satisfaction by rapid delivery of working software product.

Q. What is SRS?

Ans : Software requirements specifications is a complete description of the behaviour of the system developed. It contains a set of use cases that describes all of the interaction the users will have with software.

Q. What is oops?

Ans : Identifying objects and assigning responsibilities to objects.

Q. Write Basic Concepts of oops ?

Ans : Basic concepts of oops are as follow :

1. Class
2. Object
3. Encapsulation
4. Inheritance
5. Polymorphism
6. Abstraction

Q. What is object ?

Ans : Object is an instance of a class.

Q. What is class?

Ans : Class is a collection of data members (variables) and member function (process, method) with its behaviour.

Q. What is encapsulation?

Ans : Encapsulation is wrapping up of data into a single unit.

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Q. What is inheritance?

Ans : Inheritance is a properties of parent class extends into child class.

Q. What is polymorphism?

Ans : Polymorphism is a ability to take one name having many or multiple forms.

Q. Draw Usecase on online bill payment system (paytm)

ANS : Actor (customer)

Open app
Sign in
Select recharge and bill
Select provider
Add consumer number
Fetch bill details
Select payment mode
Password
Otp
Payment successful
Confirmation message

Q.Draw Usecase on banking system for customers.

ANS : Actor (customer)

Sign in
View Account Balance
View Transaction History
Deposit Money
Withdraw Money
Transfer Money
Add / Manage Beneficiaries
Pay Bills (electricity, mobile, etc.)
Loan
Lifestyle
Stores and offers
Forex
Social security scheme
Invest and insure
Other services
Update Profile
Logout

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Q. Draw Usecase on Broadcasting System.

ANS : Actor (customer)

Open app
Register / sign in
Watch Live Broadcast / Listen
View Program Schedule
Audio & radio broadcasting
Social & community messaging
Official and government broadcasting
Subscribe to Channels
Give Feedback / Ratings

Q. Write SDLC phases with basic introduction ?

Ans : Sdlc is a structure imposed on the development of a software production defines the process for planning, implementation, testing, documentation, deployment and ongoing maintenance support.

Sdlc phases :

Requirements gathering : gathering customer needs

Analysis : model and specify the requirements- what's

Design : ask why ?

Implementation : construct a solution in software

Testing : validate the solution against the requirements.

Maintenance : repair defects and adapt solutions to new requirements.

Q. Explain Phases of the waterfall model.

Ans :

phases of the waterfall model are follow:

1. Requirements Analysis and Specification :

This phase focuses on clearly understanding and documenting the customer's needs.

Require documented in the Software Requirement Specification (SRS), which serves as a formal agreement between the customer and the development team.

2. Design :

In this phase, the requirements from the SRS are converted into a system design that can be implemented in code.

High-Level Design (HLD):

Defines the overall system architecture, major components, and their interactions.

Low-Level Design (LLD):

Provides detailed design of each component, including logic and data flow, to guide developers during coding.

All designs are documented in the Software Design Document (SDD).

3. Development :

This phase involves converting the design into actual working software.

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Developers write source code based on the design documents.

Suitable programming languages and tools are used.

Unit testing is performed to verify that each module works correctly on its own.

4. Testing and Deployment :

This phase ensures that the integrated software functions correctly and is successfully delivered for real world use.

Testing: After unit testing, software modules are integrated incrementally and tested at each stage. Once all modules are successfully integrated, full system testing is performed to ensure the system functions as expected.

Alpha Testing: Performed by the development team.

Beta Testing: Performed by selected end users.

Acceptance Testing: Performed by the customer to approve the software.

Deployment: After successful testing, the software is deployed to a live environment for end users. This phase includes environment setup, user training, and final checks to ensure smooth operation in real world conditions.

5. Maintenance :

Maintenance ensures the software continues to function effectively after deployment.

Corrective Maintenance: Fixes errors found after release.

Perfective Maintenance: Enhances features based on user needs.

Adaptive Maintenance: Updates software to work in new environments, such as new hardware or operating systems

Q. Write phases of spiral model

Ans :

Phase of spiral model are as follow.

1. planning : determination of objectives, alternatives and constraints.
2. risk analysis : analysis of alternative and identification or resolution of risks.
3. engineering : development of the next level product.
4. Customer evaluation : assessment of the results of engineering.

Q. Write agile manifesto principles.

Ans : Agile manifesto are.

1. Individual interaction.
2. Working softwear.
3. Customer collaboration.
4. Responding to change.

Q. Explain working methodology of agile model and also write pros and cons.

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Ans : Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product.

Agile Methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about one to three weeks. Every iteration involves cross functional teams working simultaneously on various areas like planning, requirements analysis, design, coding, unit testing, and acceptance testing. At the end of the iteration a working product is displayed to the customer and important stakeholders.

Pros.

1. Is a very realistic approach to software development
2. Promotes teamwork and cross training?
3. Functionality can be developed rapidly and demonstrated.
4. Resource requirements are minimum.
5. Suitable for fixed or changing requirements
6. Delivers early partial working solutions.
7. Good model for environments that change steadily.
8. Minimal rules, documentation easily employed.
9. Enables concurrent development and delivery within an overall planned context
10. Little or no planning required Easy to manage Gives flexibility to

Cons.

1. Not suitable for handling complex dependencies.
2. More risk of sustainability, maintainability and extensibility.
3. An overall plan, an agile leader and agile PM practice is a must without which it will not work.
4. Strict delivery management dictates the scope, functionality to be delivered and adjustments to meet the deadlines.
5. Depends heavily on customer interaction, so if customer is not clear, team can be driven in the wrong direction.
6. There is very high individual dependency, since there is minimum documentation generated.
7. Transfer of technology to new team members may be quite challenging due to lack of documentation use – case

Q.Draw usecase on OTT Platform.

ANS : Actor (customer)

Viewer

Register / sign in

Search Content

Search Movies/Shows

Watch content

Add to wishlist

Download Content (offline viewing)

Rate/ comment on content

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Subscribe / Cancel Subscription
Manage Profile

Q.Draw usecase on E-commerce application

ANS : Actor (customer)

Customer
Register / sign in
Browse Products
Search Products
View Product Details
Add to Cart / Remove from Cart
Place Order
Make Payment
Track Order
Cancel / Return Order
Give Ratings & Reviews

Q.Draw usecase on Online shopping product using payment gateway.

ANS : Actor (customer)

Register / sign in
Browse & View Products
Product specifications (size,colour etc.)
Add to Cart
Place Order
Enter Payment Details
Make Payment
Track Order
View Invoice